Government Engineering College Kishanganj

Department of Computer Science and Engineering

Computer Networks LAB-03

Objective

Develop a TCP-based client-server chat application using socket programming in C.

Problem Statement

Write a C program to implement a **client-server chat application** using TCP sockets with the following requirements:

- The server must listen on **port 8081**.
- The client connects to the server using its IP address.
- Both client and server should be able to send and receive messages.
- Communication continues until the client sends the message `"bye"`.
- Upon receiving `"bye"`, the server should terminate the connection gracefully.

Additional Requirements

- Use TCP sockets (`SOCK_STREAM`).
- You may use the `string.h` library.
- Execute the program:
  - On the same machine (localhost / 127.0.0.1)
  - On different machines (using actual IP address)

Server Template

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 8081
#define BUFFER_SIZE 1024

int main() {
int server_fd, new_socket;
struct sockaddr_in address;
int addrlen = sizeof(address);
char buffer[BUFFER_SIZE];


// Create socket
server_fd = socket(AF_INET, SOCK_STREAM, 0);

// Configure address
address.sin_family = AF_INET;
address.sin_addr.s_addr = INADDR_ANY;
address.sin_port = htons(PORT);

// Bind socket

// Listen for connections

// Accept connection

// Communication loop (to be implemented)

close(new_socket);
close(server_fd);

return 0;
```

```
}
```

Client Template

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 8081
#define BUFFER_SIZE 1024

int main() {
int sock;
struct sockaddr_in serv_addr;
char buffer[BUFFER_SIZE];


// Create socket
sock = socket(AF_INET, SOCK_STREAM, 0);

// Configure server address
serv_addr.sin_family = AF_INET;
serv_addr.sin_port = htons(PORT);

// Convert IP address (127.0.0.1)

// Connect to server

// Communication loop (to be implemented)

close(sock);
return 0;


}
```